SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 7: Breadth-First Search (BFS)

CODE:

```
from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": [],
    "F": []
}

def bfs(start):
    visited = {start}
    queue = deque([start])

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbor in graph[node]:
            if neighbor not in visited:
```

```
            visited.add(neighbor)
            queue.append(neighbor)

bfs("A")
```

OUTPUT:

